

API Reference

Comprehensive documentation for the Gema Event Management Platform API.

Quick Navigation

Section	Description	Authentication
Public APIs (public-apis.md)	Browse events, categories, and public data	None required
Authentication (./authentication.md)	User registration, login, and token management	Public + Private
User Management (user-management.md)	Profile management and user preferences	 User auth
Events (events.md)	Event creation, management, and discovery	 Vendor/Admin
Orders & Payments (orders-payments.md)	Order processing and payment handling	 Customer auth
Venues & Vendors (venues-vendors.md)	Venue and vendor management	 Vendor/Admin
Reviews & Ratings (reviews-ratings.md)	Review system and moderation	 Customer auth
File Uploads (file-uploads.md)	Image and document management	 Auth required
Admin APIs (admin-apis.md)	Administrative functions	 Admin only

Base URLs

Environment	URL	Status
Development	http://localhost:5000/api	☒ Local development
Staging	https://gema-backend-staging.render.com/api	 Testing environment
Production	https://gema-backend.render.com/api	 Live system

Authentication Overview

Most endpoints require authentication via JWT tokens:

```
# Include in request headers
Authorization: Bearer <your_access_token>
Content-Type: application/json
```

Token Types:

- **Access Token:** Short-lived (7 days), used for API requests
- **Refresh Token:** Long-lived (30 days), used to obtain new access tokens

Response Format

All API responses follow a consistent structure:

Success Response

```
{
  "success": true,
  "message": "Operation completed successfully",
  "data": {
    // Response data here
  }
}
```

Error Response

```
{
  "success": false,
  "message": "Error description",
  "error": {
    "code": "ERROR_CODE",
    "details": "Additional error information"
  }
}
```

Paginated Response

```
{
  "success": true,
  "message": "Data retrieved successfully",
  "data": {
    "items": [...],
    "pagination": {
      "currentPage": 1,
      "totalPages": 10,
      "totalItems": 100,
      "hasNextPage": true,
      "hasPrevPage": false,
      "limit": 10
    }
  }
}
```

HTTP Status Codes

Code	Meaning	Description
200	OK	Request successful
201	Created	Resource created successfully
400	Bad Request	Invalid request data/validation failed
401	Unauthorized	Authentication required or token invalid
403	Forbidden	Insufficient permissions for this action
404	Not Found	Requested resource doesn't exist
409	Conflict	Resource already exists (duplicate)
422	Unprocessable Entity	Request valid but cannot be processed
429	Too Many Requests	Rate limit exceeded
500	Internal Server Error	Unexpected server error

Rate Limiting

API requests are rate-limited to prevent abuse:

- **Default Limit:** 100 requests per 15-minute window
- **Headers Included:**
 - X-RateLimit-Limit: Maximum requests allowed
 - X-RateLimit-Remaining: Remaining requests in current window
 - X-RateLimit-Reset: Time when the rate limit resets

Pagination

List endpoints support pagination with consistent parameters:

Query Parameters

- **page (number):** Page number (default: 1)
- **limit (number):** Items per page (varies by endpoint, max: 50-100)
- **sortBy (string):** Field to sort by (varies by endpoint)
- **sortOrder (string):** "asc" or "desc" (default: "desc")

Example Request

```
GET /api/events?page=2&limit=20&sortBy=createdAt&sortOrder=desc
```

Filtering & Search

Many endpoints support filtering and search:

Common Filters

- **Text Search:** `search` parameter for full-text search
- **Date Ranges:** `dateFrom`, `dateTo` parameters
- **Status Filters:** `status` parameter with predefined values
- **Category Filters:** `category`, `type` parameters

Example Request

```
GET /api/events?search=music&category=Entertainment&dateFrom=2024-03-01&dateTo=2024-03-31
```

Security Best Practices

Authentication

1. **Store tokens securely** - Use secure storage mechanisms
2. **Refresh tokens before expiry** - Implement automatic token refresh
3. **Don't expose sensitive data** - Never log or expose tokens

Request Security

1. **Use HTTPS in production** - All production requests must use HTTPS
2. **Validate input** - Client-side validation doesn't replace server-side
3. **Handle errors gracefully** - Don't expose internal system details

Getting Started

1. **New to the API?** Start with [Getting Started Guide \(../getting-started.md\)](#)
2. **Need examples?** Check out [Examples Directory \(../examples/\)](#)
3. **Having issues?** See [Troubleshooting Guide \(../troubleshooting.md\)](#)
4. **Ready to develop?** Read [Development Guidelines \(../development.md\)](#)

Additional Resources

- [Authentication Deep Dive \(../authentication.md\)](#) - Detailed auth implementation
- [Deployment Guide \(../deployment.md\)](#) - Production deployment steps
- [Code Examples \(../examples/\)](#) - Common workflows and use cases
- [Troubleshooting \(../troubleshooting.md\)](#) - Common issues and solutions

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